The Curious Case of MuonSign and SignMuon: Sign Before or After the LMO?

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Abstract

The SignSGD gradient compression algorithm enables up to a $32\times$ reduction in communication volume, which is critical for federated learning in bandwidth-constrained environments. It often lags behind modern optimizers like Muon, which leverage the matrix structure of parameters to achieve superior convergence and performance. We propose SignMuon, an algorithm that applies sign compression to the Linear Minimization Oracle (LMO) update of the Muon optimizer, i.e., it computes $\mathrm{sign}(\mathrm{LMO}(\cdot))$. However, SignMuon offers no global convergence guarantee: we construct a linear function on which SignMuon provably diverges. Based on this observation, we introduce EF21-MuonUSign, an Error-Feedback variant that compresses the residual with $\mathrm{sign}(\cdot)$ and applies the Muon LMO on the server, computing $\mathrm{LMO}(\mathrm{sign}(\cdot))$ while keeping the uplink at one bit per parameter. We further introduce EF21-MuonUDSign, a bidirectional variant that additionally compresses the downlink with the scaled-sign operator, reducing communication to one bit per parameter in both directions while retaining convergence guarantees. It restores the descent property and converges on the same counterexample. We validate both methods on a synthetic convex problem and on centralized and federated CIFAR-10 classification.

Introduction

Modern deep neural network (DNN) training methods require optimizers that not only achieve fast convergence but also minimize communication overhead in distributed systems. In federated learning, data are distributed across clients that communicate with a central server to train a global model. Each client possesses its own dataset and updates its parameters using a local optimizer. The goal of federated learning is to minimize the average loss function across all clients. Under limited network bandwidth, transmitting full model updates can significantly slow down the training process. Gradient compression methods are commonly used to address this issue (Bernstein et al. 2018, 2019; Beznosikov et al. 2024).

For instance, the SignSGD algorithm has demonstrated that even extreme compression (down to 1 bit per component) can preserve the practical effectiveness of stochastic optimization while reducing the communication volume. This data volume is reduced by a factor of approximately $32\times$ compared to standard SGD, with only minor degradation in model quality (Bernstein et al. 2018, 2019). However, SignSGD does not account for the matrix structure of the parameters. In tasks with matrix weights, it treats matrices as one-dimensional vectors (flattening), ignoring their intrinsic two-dimensional tensor structure, which may negatively affect the training performance of DNNs.

Recently, Muon has demonstrated that orthonormalized updates can stabilize DNN training and, in several cases, outperform standard adaptive optimization methods in terms of final solution accuracy (Jordan et al. 2024; Bernstein and Newhouse 2024; Essential AI 2025). For matrix parameters, Muon implements a steepest descent step in the spectral norm. According to Jordan et al. (2024), the update direction is computed as the orthonormal matrix $\mathbf{U}\mathbf{V}^\top$, where $\mathbf{U}$ and $\mathbf{V}$ are the singular-vector matrices obtained from the gradient decomposition. Based on the structure of the algorithm, Muon requires the transmission of full update matrices, which can be prohibitively computationally expensive for federated tasks.

We propose **SignMuon**, which applies sign compression to the Muon LMO direction: it preserves the matrix-aware geometry of Muon while communicating only one bit per parameter. Empirically, SignMuon matches Muon's accuracy and significantly outperforms SignSGD in both centralized and federated learning.

However, this combination has no global convergence guarantee. We prove (Theorem 1) that SignMuon can diverge: on a 4×4 counterexample, the sign-compressed LMO step becomes an ascent direction. To restore convergence without increasing the communication volume, we introduce **EF21-MuonUSign**, an Error-Feedback variant adapted from EF21-Muon Gruntkowska et al. (2025). It reconstructs the gradient through a contractive sign compressor and applies the Muon LMO on the server. EF21-MuonUSign converges on the same counterexample while retaining uplink one-bit communication.

Our experiments investigate how effectively Muon-style structured update directions can be combined with the extreme communication compression characteristic of sign-based methods. Specifically, we analyze how transmitting only the update direction signal affects the accuracy and stability of training compared to using the full Muon update. We validate both methods on centralized image classification on CIFAR-10 with a ResNet-18 backbone, federated image classification on CIFAR-10 with a CNN across several client counts, and on a synthetic smooth convex least-squares problem. Our contributions are: (i) SignMuon, a matrix-aware 1-bit optimizer; (ii) a counterexample (Theorem 1) proving SignMuon may diverge; (iii) EF21-MuonUSign, which compresses the uplink to one bit per parameter while restoring convergence guarantees; (iv) EF21-MuonUDSign, a bidirectional variant that adds scaled-sign compression on the downlink, keeping both channels at one bit per parameter.

Related Work

SignSGD. Vanilla SignSGD, introduced by Bernstein et al. (2018) and later applied to federated learning by Bernstein et al. (2019), required increasing batch sizes to guarantee convergence. This issue, together with SignSGD's inherent bias, was addressed through the adoption of the Error Feedback technique (Karimireddy et al. 2019). An alternative approach is to randomize the sign operator itself (Chen et al. 2020; Safaryan and Richtarik 2021; Jin et al. 2025).

Compression for Muon. Gruntkowska et al. (2025) developed a theoretical framework for federated learning with error feedback in which Muon/Gluon (Riabinin et al. 2025) updates are compressed bidirectionally (client-to-server and server-to-client). Qian et al. (2026) employed variance reduction to lower the compression error for Gluon under unbiased and contractive compressors. Concurrently with our work, Mishra, Trivedi, and Kumar (2026) proposed vanilla SignMuon and demonstrated its empirical feasibility, but did not identify an instance of SignMuon's non-convergence such as the one given in Theorem 1.

Problem Statement

Centralized Learning: We consider the stochastic optimization problem:

$$\min_{\mathbf{X}\in\mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f_\xi(\mathbf{X})]\}, \tag{1}$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$), functions $f_\xi : \mathcal{X} \to \mathbb{R}$ are continuously differentiable (possibly non-convex). Here, $f_\xi(\mathbf{X})$ is the loss function of a model parameterized by $\mathbf{X}$, computed at a data point ξ, sampled from the probability distribution $\mathcal{D}$.

Federated Learning: In modern machine learning, progress is increasingly driven by the ability to scale computational systems. SOTA models rely on massive datasets and highly complex architectures, often requiring substantial computational resources and extended training times. As a result, the training process is formulated as an optimization problem in a distributed environment:

$$\min_{\mathbf{X}\in\mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N}\sum_{j=1}^{N} f_j(\mathbf{X}) \right\},$$
$$f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X};\xi_j)] \tag{2}$$

where $\mathbf{X} \in \mathcal{X}$ are the model parameters, $N \geq 1$ is the number of clients, and $f_j(\mathbf{X})$ is the loss of the model $\mathbf{X}$ on the data $\mathcal{D}_j$, stored on client $j \in [N] := \{1,\dots,N\}$. In federated learning, clients do local computations and synchronize with a global server. Since communication bandwidth is often limited, the volume of transmitted data becomes a major bottleneck in the training process (Bernstein et al. 2019; Horváth et al. 2019). Several practical techniques for federated optimization and communication compression have also been studied in (Beznosikov et al. 2024; Gruntkowska et al. 2025; Ahn et al. 2025).

For theoretical optimization analysis in distributed environments, we rely on Assumption 2 that the objective function f is smooth w.r.t. the spectral norm. We note that, in practice, weaker smoothness assumptions such as the generalized (L_0, L_1)-smoothness condition may also be considered (Pethick et al. 2026). Limited network bandwidth makes the transmission of full-precision 32-bit gradients prohibitively expensive, especially when training large-scale models. This motivates the study of communication-efficient optimization methods based on extreme gradient compression (Bernstein et al. 2018, 2019; Beznosikov et al. 2024).

Linear Minimization Oracle (LMO) is defined as an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set. In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\{\mathbf{Y}\in\mathcal{S}\mid\|\mathbf{Y}\|\leq 1\}} \langle \mathbf{X}, \mathbf{Y}\rangle, \tag{3}$$

where $\mathbf{X} \in \mathbb{R}^{m\times n}$ is the input matrix, $\mathcal{S} \subset \mathbb{R}^{m\times n}$ is a convex set, and $\langle \mathbf{X}, \mathbf{Y}\rangle = \sum_{i,j} X_{ij}Y_{ij}$ is the standard matrix inner product (Pethick et al. 2025).

Such an operator selects the direction of steepest descent with respect to the specified norm. The analytical and practical role of the LMO is discussed in detail in the literature on norm-constrained LMO optimizers (Pethick et al. 2025; Kovalev 2025; Riabinin et al. 2025).

Muon optimizer uses the matrix structure of the gradient. Muon implements a steepest descent step in the spectral norm, where the update direction is obtained by orthogonalizing the gradient matrix (Bernstein and Newhouse 2024). Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \mathrm{rank}(\mathbf{M})$. Then, the LMO direction takes the following form:

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m\times n}. \tag{4}$$

That is, Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. In practice, Muon utilizes a smoothed gradient estimate with momentum, and its update is defined as follows:

$$\begin{aligned}\mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t,\\ \mathbf{M}_t &= \mathbf{U}_t \mathbf{\Sigma}_t \mathbf{V}_t^\top,\\ \mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t) = \mathbf{U}_t \mathbf{V}_t^\top\\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \mathbf{D}_t.\end{aligned} \tag{5}$$

In the original Muon implementation, the computation of $\mathbf{U}_t \mathbf{V}_t^\top$ is approximated using Newton-Schulz iterations and related polynomial schemes. It enables efficient matrix orthogonalization without an explicit SVD computation (Amsel et al. 2026; Li and Hong 2025; Liu et al. 2025). Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN (Bernstein and Newhouse 2024; Essential AI 2025).

Despite its advantages, Muon requires transmitting the full update matrix $\mathbf{D}_t$, making it expensive in federated settings. Our objective is to combine Muon's matrix-aware geometry with extreme (1-bit) communication compression. SignMuon achieves this combination empirically, but, as we show (Theorem 1), it lacks global convergence guarantees. The Error-Feedback variant EF21-MuonUSign restores convergence: the EF21 estimator bounds established in Appendix provide the key ingredients for the standard $\mathcal{O}(T^{-1/2})$ rate for smooth non-convex optimization (Hypothesis 1):

$$\min_{0 \le t \le T} \mathbb{E}\left[\|\nabla f(\mathbf{X}_t)\|_*^2\right] = \mathcal{O}(T^{-1/2}). \tag{6}$$

Theory

Consider the SignA family of algorithms, where A denotes any optimizer based on LMO directions. All methods in this family follow the same three-stage structure: momentum accumulation, computation of an LMO direction, and sign-compression of the resulting update.

Formally, a SignA method initializes the momentum buffer as $\mathbf{M}_0 = \mathbf{0}$, and at each iteration $t \ge 1$ performs the following sequence of updates:

$$\begin{aligned}
\mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t && \text{(Momentum accumulation)}\\
\tilde{\mathbf{M}}_t &= \begin{cases}\mathbf{G}_t + \mu \mathbf{M}_t & \text{(Nesterov momentum)},\\ \mathbf{M}_t & \text{(otherwise)}\end{cases} && \text{(Effective direction)},\\
\mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t), && \text{(LMO direction)}\\
\mathbf{s}_t &= \operatorname{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, && \text{(Sign compression)}\\
\mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \lambda_{\text{mult}} \mathbf{s}_t && \text{(Parameter update)},
\end{aligned} \tag{7}$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the current model parameter matrix, $\mathbf{G}_t$ is its stochastic gradient, $\mu \in [0, 1)$ is the momentum coefficient, η_t is the learning rate, and $\lambda_{\text{mult}} > 0$ is an optional scaling factor (with $\lambda_{\text{mult}} = 1$ by default). The operator $A(\cdot)$ denotes a Linear Minimization Oracle (LMO) in a specified non-Euclidean norm, and the function $\operatorname{sign}(\cdot)$ is applied element-wise to the resulting structured direction $\mathbf{D}_t$.

The proposed SignMuon algorithm is a particular instance of the SignA family. It combines the orthogonalized LMO directions of the Muon optimizer with the extreme sign-based compression strategy of SignSGD. In the following sections, we present SignMuon and EF21-MuonUSign, each in centralized and federated variants.

Centralized Learning

In the centralized setting, SignMuon is applied to the matrix-valued parameters (weights of convolutional and fully connected layers, $n_{dim} \ge 2$), while vector parameters (biases, BatchNorm, $n_{dim} = 1$) and the final classification layer are optimized with AdamW – the design introduced by Jordan et al. (2024) and formalized in (Li and Hong 2025; Riabinin et al. 2025).

For matrix parameters, the key component of SignMuon is the computation of an orthonormal LMO direction $\mathbf{D}_t \approx \mathbf{U}_t \mathbf{V}_t^\top$. To avoid the computational cost of performing a full singular value decomposition (SVD) at every optimization step, the orthogonalization is approximated using a 5th-order Newton-Schulz iteration (Algorithm 1). In our implementation, the Newton-Schulz coefficients are chosen as $a = 3.4445, b = 4.7750, c = 2.0315$. The main difference between SignMuon and the classical SignSGD algorithm lies in the object being compressed. In SignSGD, sign compression is applied directly to the stochastic gradient. In contrast, SignMuon first constructs a structured update direction $\mathbf{D}_t$, corresponding to the Muon LMO step and only then applies the elementwise sign operator to obtain $\operatorname{sign}(\mathbf{D}_t)$. Consequently, SignMuon preserves the geometry of orthonormal updates while reducing the transmitted information to one bit per component (Algorithm in Appendix).

Federated Learning

In federated learning, the training data are distributed across N clients. Each client j possesses a local dataset $\mathcal{D}_j$ and computes a local loss function $f_j(\mathbf{X})$. The goal of the training process is to minimize the averaged loss function across all clients (2). Since clients do not exchange raw data and communicate only through messages sent to the server, communication efficiency becomes a critical concern in federated optimization.

In the federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t, each client j receives a copy of the current global model. Rather than performing local parameter updates, the client computes an averaged stochastic gradient by accumulating gradients over $\mathrm{n\_steps}$ local mini-batches. This gradient accumulation procedure reduces the variance of the stochastic gradient estimate before the application of the LMO operator. Formally, the accumulated gradient is defined as:

$$\mathbf{G}_t^{(j)} \leftarrow \frac{1}{n_{\text{steps}}} \sum_{i=1}^{n_{\text{steps}}} \nabla f_j(\mathbf{X}_t; \xi_{t,i}^{(j)}).$$

In accordance with the general SignA framework (7), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the accumulated gradient, the client updates the momentum estimate, applies the LMO (implemented via

Newton-Schulz orthogonalization, Algorithm 1), and then performs element-wise sign compression:

$$\mathbf{M}_t^{(j)} = \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$$

$$\mathbf{s}_t^{(j)} = \mathrm{sign}\big(A(\mathbf{M}_t^{(j)})\big) \in \{\pm 1\}^{m \times n}$$

where $t \in [0, \mathrm{rounds} - 1]$, $A(\cdot)$ is the Muon LMO (Newton-Schulz, as in the centralized setting). Thus, each client transmits only 1 bit per matrix parameter component to the server.

On the server, messages received from the clients are aggregated using a majority vote. The aggregated sign is denoted by $\mathbf{s}_t^{\mathrm{agg}}$, then it is given by the formula:

$$\mathbf{s}_t^{\mathrm{agg}} = \mathrm{sign}\left(\sum_{j=1}^{N} \mathbf{s}_t^{(j)}\right). \tag{8}$$

The majority-vote operation ensures that the resulting sign $\mathbf{s}_t^{\mathrm{agg}}$ is equal to $+1$ or -1 in each component and corresponds to the "majority vote" of the clients. Since momentum has already been taken into account at the clients' level, the server directly updates the matrix parameters of the global model:

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\mathrm{agg}}.$$

A special treatment is applied to the final classification layer of the model. These parameters are not updated using the SignMuon rule; instead, they are optimized using AdamW. The complete federated SignMuon procedure is presented in Appendix.

Theoretical limitations of SignMuon

The hypotheses presented in Appendix rely on the assumption that the direction of steepest descent in the spectral norm (the LMO step of the Muon algorithm) remains a valid descent direction even after applying extreme sign compression. However, this is not always the case.

The classic gradient descent algorithm guarantees that the anti-gradient direction $(-\mathbf{G}_t)$ is a descent direction. In the SignMuon algorithm, the step direction is defined by the matrix $\mathbf{s}_t = \mathrm{sign}(\mathbf{U}_t \mathbf{V}_t^\top)$. For the algorithm to guarantee convergence, the following descent condition must be met:

$$\langle \mathbf{G}_t, \mathrm{sign}(\mathbf{U}_t \mathbf{V}_t^\top) \rangle > 0. \tag{9}$$

This raises the question: does spectral orthogonalization guarantee that condition (9) holds for arbitrary matrices? We formalize this question in the following theorem.

Theorem 1 (Divergence counterexample) *There exists a smooth function $f(\mathbf{W})$ and a matrix dimension $N_{dim} \geq 4$ such that the SignMuon algorithm diverges.*

A proof by counterexample is provided in Appendix. This counterexample demonstrates that the SignMuon algorithm does not guarantee global convergence for general smooth functions. This motivates the development of a variant with Error Feedback, which preserves one-bit uplink communication while restoring convergence.

Centralized Learning with Error Feedback

We define the **USign** compressor as the directional pair $\mathrm{USign} := (\mathcal{C}_\uparrow, \mathcal{C}_\downarrow)$, where $\mathcal{C}_\uparrow(\mathbf{Y}) = \mathrm{mean}\,|\mathbf{Y}|\,\mathrm{sign}(\mathbf{Y})$ is the scaled-sign operator on the uplink (client → server) and $\mathcal{C}_\downarrow = I$ is the identity on the downlink (server → client). Unlike a fully 1-bit scheme, USign compresses only the uplink to one bit per parameter, while the server broadcasts the full model. **EF21-MuonUSign** pairs USign with EF21 and the Muon LMO: the sign acts on the EF21 residual, and the optimization step uses the full Muon LMO of the reconstructed estimator.

Theorem 1 shows that sign-compressing the LMO direction can destroy the descent property. To restore convergence without increasing the communication volume, we replace the sign-compressed step $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t\,\mathrm{sign}(\mathbf{D}_t)$ by an Error-Feedback variant inspired by EF21-Muon (Gruntkowska et al. 2025).

EF21-MuonUSign maintains a gradient estimator $\mathbf{g}_t^{\mathrm{est}}$ updated via a contractive sign compressor on the residual $\Delta_t = \mathbf{M}_t - \mathbf{g}_t^{\mathrm{est}}$:

$$\alpha_t = \mathrm{mean}(|\Delta_t|), \quad \mathbf{g}_{t+1}^{\mathrm{est}} = \mathbf{g}_t^{\mathrm{est}} + \alpha_t\,\mathrm{sign}(\Delta_t), \tag{10}$$

and takes the descent step

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t\,\lambda_{\mathrm{mult}}\,\mathbf{D}_t, \qquad \mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\mathrm{est}}) \approx \mathbf{U}_{t+1}\mathbf{V}_{t+1}^\top, \tag{11}$$

As $\mathbf{g}_t^{\mathrm{est}} \to \mathbf{M}_t$, the direction $\mathbf{D}_t \to \mathbf{U}_t \mathbf{V}_t^\top$ is always a descent direction, since $\langle \nabla f,\, \mathbf{U}_t\mathbf{V}_t^\top\rangle \approx \|\nabla f\|_* > 0$. The sign in (10) acts only on the internal compression residual, keeping the uplink at one bit per parameter. The Algorithm 4 is given in Appendix. Figure 1 confirms that EF21-MuonUSign converges on the same counterexample where SignMuon diverges. We now extend this Error-Feedback mechanism to the federated setting.

While USign leaves the downlink uncompressed, a fully 1-bit scheme is obtained by compressing both channels. We define the **UDSign** compressor as the directional pair $\mathrm{UDSign} := (\mathcal{C}_\uparrow, \mathcal{C}_\downarrow)$ with the same scaled-sign operator on both channels, $\mathcal{C}_\uparrow(\mathbf{Y}) = \mathcal{C}_\downarrow(\mathbf{Y}) = \mathrm{mean}\,|\mathbf{Y}|\,\mathrm{sign}(\mathbf{Y})$. In this version of Error Feedback Algorithm the node keeps the exact model $\mathbf{X}_t$ and takes the exact LMO step, while it observes only a compressed model shift $\mathbf{W}_t$ reconstructed by error feedback,

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \lambda_{\mathrm{mult}} \mathbf{D}_t, \qquad \mathbf{W}_{t+1} = \mathbf{W}_t + \alpha_t^\downarrow\,\mathrm{sign}(\mathbf{X}_{t+1} - \mathbf{W}_t), \tag{12}$$

with the gradient evaluated at $\mathbf{W}_t$ and $\mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\mathrm{est}})$. The full procedure is in Algorithm 5.

It should be noted that both error feedback methods require additional memory: each compressed channel maintains its own error feedback buffer (the gradient estimate $\mathbf{g}^{\mathrm{est}}$ and the model shift buffer $\mathbf{W}$ in the case of bidirectional feedback). This additional overhead is low in terms of server memory, but may be unwanted on worker nodes with limited memory.

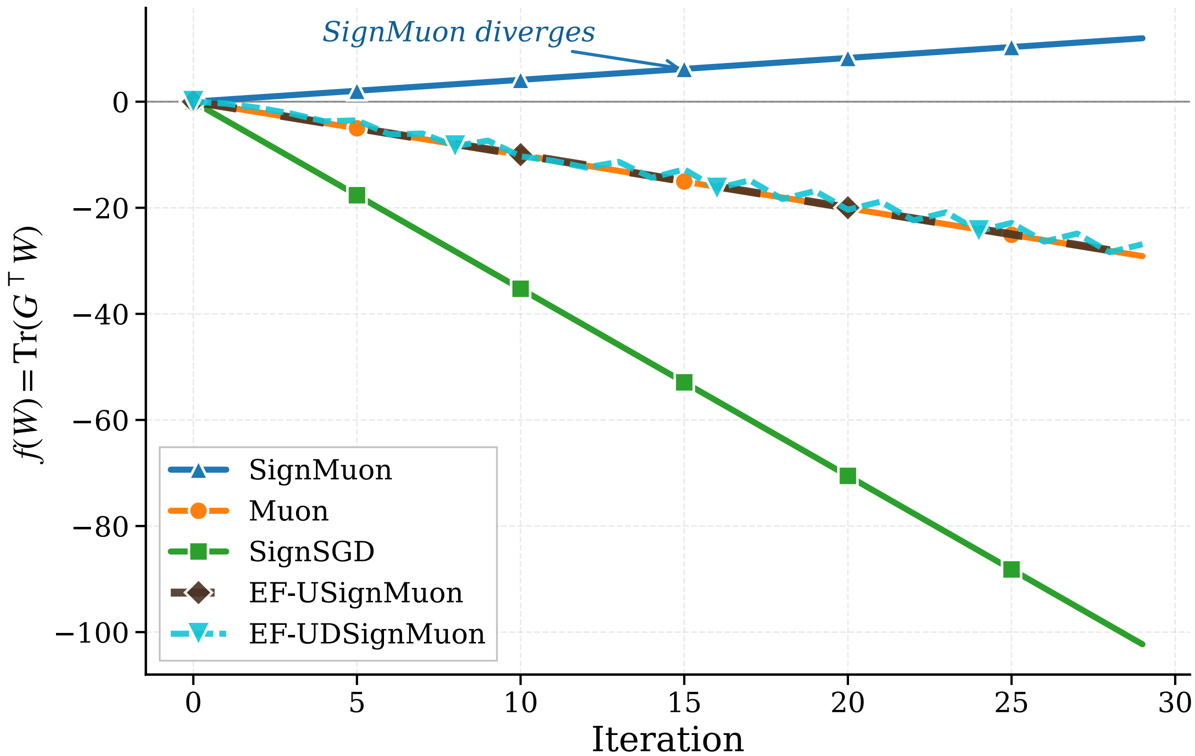


Figure 1: Comparison of optimization trajectories for the linear objective function $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W})$, demonstrating the strict divergence of SignMuon, while other methods successfully minimize the function.

Federated Learning with Error Feedback

Although Federated SignMuon demonstrates strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for quantization error and ensure strong convergence guarantees, we incorporate an Error Feedback method. In the federated setting, the Error-Feedback mechanism is adapted from EF21-Muon (Gruntkowska et al. 2025).

The LMO orthogonalization is moved from the clients to the server: each client evaluates the difference between its local momentum $\mathbf{M}_t^{(j)}$ and the previous gradient estimate $\mathbf{g}_t^{(j)}$, compresses it via scaled sign, and sends only $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$. The server aggregates these, updates a global gradient estimator $\mathbf{G}_{t+1}$, and applies the Muon LMO. The complete Algorithm 7 is in Appendix. The transmission of an additional scalar $\alpha_t^{(j)}$ for each layer introduces insignificant communication overhead, keeping the uplink communication cost at ≈ 1 bit per parameter (the server broadcasts the full model on the downlink). Thus, both federated variants – SignMuon and EF21-MuonUSign – effectively address: the baseline majority-vote scheme and the error-feedback variant – effectively address the communication bottleneck in federated learning. They preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

Experiments

The goal of these experiments is empirical validation of the SignMuon and EF21-MuonUSign algorithms. The study consists of several parts, including evaluation in both centralized and federated settings, as well as validation on a synthetic least-squares optimization problem.

Smooth Convex Optimization Problem (Synthetic Experiment)

To evaluate the effect of matrix structure on the optimization process (without the influence of stochastic noise or the complexity of DNN architectures), we begin our experiments with a deterministic L-smooth convex optimization problem. We consider the minimization of a quadratic form:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B}\rangle \to \min_{\mathbf{X}\in\mathbb{R}^{m\times n}}, \quad (13)$$

where the parameter matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$ has dimension $m = n = 500$, which is representative of the size of parameter matrices in deep neural networks. The matrices $\mathbf{A} \in \mathbb{S}_+^m$ and $\mathbf{B} \in \mathbb{S}_+^n$ are symmetric positive semidefinite. To generate the problem instance, we randomly construct the spectra of $\mathbf{A}$ and $\mathbf{B}$ by sampling their eigenvalues uniformly from the interval $(0, 1)$, which guarantees a smoothness constant $L \leq 1$ with respect to the Frobenius norm. The parameter matrix

Algorithm	Learning rate (η)	Momentum (μ)	Iterations to 0.001
SignMuon	0.0002	0.2	2104
Muon	0.0065	0.1	1122
EF21-MuonUSign	0.0033	0.1	3694
SignSGD	0.00015	0.8	3120
SGD	0.1	0.95	972
Adam	0.07	-	202

Table 1: Tuned learning rates and momentum coefficients in the experiments on the smooth convex optimization problem (Equation 13).

$\mathbf{X}_0$ is initialized with elements sampled independently from a normal distribution $\mathcal{N}(0, 0.01)$.

Since theoretical convergence guarantees depend on the choice of norm and the corresponding smoothness constants, we adopt an empirical evaluation criterion. Specifically, the optimizers are compared by the minimum number of iterations required to reach the target function $F(\mathbf{X}) \leq 0.001$, with a maximum number of iterations set to $T_{\max} = 5000$. We perform a grid search over the hyperparameter ranges for each of the optimization algorithms under consideration (Table 3).

The results of the grid search with the number of iterations required to converge to 0.001, are summarized in Table 1. Muon converges in 1122 iterations, comparable to SGD (972) but slower than Adam (202). SignMuon reaches the target in 2104 iterations, significantly outperforming SignSGD (3120). EF21-MuonUSign requires 3694 iterations; its overhead on this benign convex problem is expected, since its purpose is to guarantee convergence on problems where SignMuon does not converge (Theorem 1). The optimization trajectories are shown in Appendix, Figure 3.

Centralized Learning

In this experiment, we compare the convergence dynamics and the final accuracies of SignMuon and EF21-MuonUSign with those of the classical optimizers: Muon, SignSGD, SGD, and Adam at a fixed number of epochs. One of the goals of the experiment is to demonstrate that SignMuon achieves classification quality comparable to Muon, while outperforming the classical SignSGD algorithm under the same uplink communication budget (1 bit per parameter on the uplink).

The experiments are conducted on CIFAR-10. It is a standard image classification dataset. It contains 60,000 color images of 10 classes. Each image is represented as a tensor of size $3 \times 32 \times 32$, where the first dimension corresponds to the number of color channels (RGB). For centralized training, the dataset is split into training and test sets. In federated learning, the training set is partitioned across the clients.

The base model is a ResNet-18 architecture adapted to low-resolution images by using an initial 3×3 convolution without max-pooling. The network consists of 4 residual blocks with Batch Normalization (BatchNorm), followed by a dropout regularization layer and a linear classifier. As mentioned above, training is run for a fixed number of epochs $T_{\max} = 75$. During training, the learning rate is annealed with a cosine scheduler (CosineAnnealingLR). The learning rate η_t at epoch t is updated as follows:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T_{\max}}\pi\right)\right), \quad (14)$$

where $\eta_{\max}$ is the initial learning rate of the algorithm, $\eta_{\min} = 0$ is the minimum value (in our configuration), and $T_{\max}$ is the total number of training epochs.

The experimental results and the selected hyperparameter values are reported in Appendix in Table 4. The dynamics of the loss function and of the accuracy of the algorithms during training are illustrated in Figure 2. Notably, EF21-MuonUSign achieves 93.82% test accuracy, surpassing SignMuon (92.55%) and matching Muon (93.70%), confirming that the Error-Feedback mechanism does not sacrifice practical accuracy.

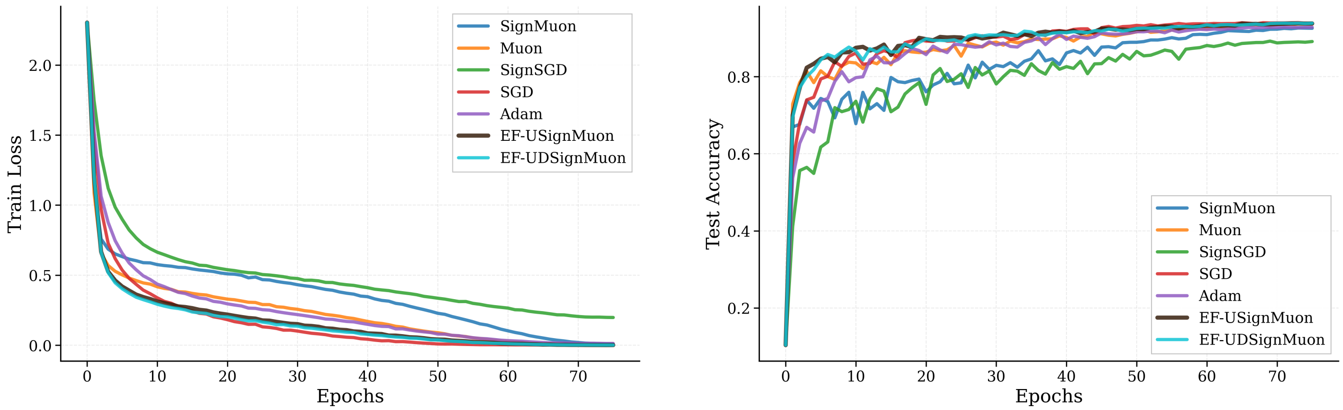
Federated Learning

The next stage of the research was adaptation of SignMuon in federated learning, where transmitting full 32-bit gradient matrices is the communication bottleneck. The goal of this stage is to confirm that extreme compression of the orthogonalized updates attains an accuracy comparable to that of Muon, while significantly reducing the volume of transmitted data.

For these experiments, we implemented a federated architecture with a central aggregation server. To keep the results comparable with the previous stages, we used the convolutional neural network (CNN2), comprising two convolutional blocks with BatchNorm and an MLP classifier. The CIFAR-10 dataset was partitioned across the clients homogeneously (homogeneous split). Within this learning type, we study both the baseline Federated SignMuon scheme (Algorithm 6) and the error-feedback variant Federated EF21-MuonUSign (Algorithm 7). Each communication round consists of a local gradient accumulation step on the clients, followed by transmitting the sign-compressed updates to the server, which updates the global model.

For a systematic analysis of the proposed algorithms, this experimental stage is split into several sub-experiments. First, we assess how the scale of the federated network affects the stability of sign-based optimization and the final accuracy of the model. To this end, we study the baseline SignMuon algorithm for different numbers of active clients.

Experiment 1: Scalability of SignMuon with respect to the number of clients. We investigate the effect of the



(a) Train Loss

(b) Test Accuracy

Figure 2: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms while training a ResNet-18 classifier on CIFAR-10.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	10	3	64	0.001	0.9	84.57%
	Muon					0.05		85.22%
	SignSGD					0.001		78.98%
	SGD					0.03		81.38%
	Adam					0.001		78.36%
	EF21-MuonUSign					0.02		84.59%

Table 2: CIFAR-10: federated learning. Experiment 3: Comparison of the algorithms with $N = 10$ clients.

number of participating clients on the training quality of SignMuon. The number of clients is $N \in \{1, 3, 5, 7, 9\}$, with 2000 training rounds and one local step per client. The results show that classification accuracy grows with the number of clients. This effect is explained by the improved quality of majority-vote aggregation as more independent sign estimates of the gradient become available. The results of this experiment are reported in Appendix.

Having confirmed the positive effect of the number of clients on the stability of the algorithm, we proceed to a direct comparison of SignMuon with the existing algorithms. For this purpose, we select a configuration with 3 clients, which allows us to evaluate the efficiency of the algorithms under limited federation resources.

Experiment 2: Comparison of the algorithms with $N = 3$ clients. We compare 6 optimizers with 3 clients, 2000 rounds, and 1 local step. The results of this experiment are reported in Appendix.

However, the potential of sign-based optimization algorithms is most fully revealed when the scale of the network and the amount of local computation increase. Therefore, at the final testing stage we increased the number of clients to 10.

Experiment 3: Comparison of the algorithms with $N = 10$ clients. The same 6 optimizers are tested with 10 clients, 2000 rounds, and 3 local steps per client. In this learning type, SignMuon reaches the quality of Muon. We conclude that, with a sufficient number of clients ($N \geq 10$), SignMuon achieves a quality comparable to Muon while reducing the volume of transmitted data by a factor of $\sim$32×, by transmitting only the signs of the gradients instead of their full values. The results of this experiment are reported in Table 2 and Figure 6. EF21-MuonUSign (84.59%) performs on par with SignMuon (84.57%), demonstrating that the convergence guarantee comes at no accuracy cost in the federated setting.

Conclusion

This work proposes SignMuon, which applies 1-bit sign compression to the Muon LMO direction to bridge structured optimization and extreme communication efficiency in federated learning. While it empirically matches Muon's accuracy and outperforms SignSGD, we formally prove that direct sign-compressed LMO lacks global convergence guarantees and can diverge. To resolve this without compromising uplink efficiency, we introduce EF21-MuonUSign, an Error-Feedback variant that applies sign compression to the internal residual rather than the LMO step. EF21-MuonUSign restores convergence on our counterexample and achieves Muon-level performance while maintaining a 1-bit uplink. Our framework establishes a foundation for adapting matrix-aware optimizers to bandwidth-constrained distributed training without sacrificing model accuracy.

Future work. An extension of EF21-MuonUSign is a bidirectional scheme that compresses both the uplink (client →

server) and the downlink (server $\to$ client), in the spirit of the two-sided framework of Gruntkowska et al. (2025). In this work we focus on uplink compression, since the aggregation of many client messages is typically the dominant communication bottleneck, and we keep the downlink at full precision ($\mathcal{C}_\downarrow = I$). Furthermore, investigating how these matrix-aware 1-bit strategies scale to the massive parameter matrices of Large Language Models is a highly promising direction for future research.

Appendix

Proof of Theorem 1 (Divergence counterexample)

Proof The proof is constructed using a counterexample that shows the existence of a smooth function and a matrix dimension $N_{dim} \geq 4$ for which the SignMuon algorithm is guaranteed to diverge. We conjecture, supported by extensive numerical search, that no such counterexample exists for $N_{dim} \leq 3$. However, at $N_{dim} = 4$, this convergence guarantee vanishes.

Consider a simple linear objective function $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W})$. For standard gradient descent or Muon, such a function diverges towards $-\infty$. The gradient is globally constant: $\nabla f(\mathbf{W}_t) = \mathbf{G}$ for all t. The change in the objective function at each iteration of the SignMuon algorithm is given by:

$$f(\mathbf{W}_{t+1}) - f(\mathbf{W}_t) = -\eta\langle \mathbf{G}, \mathrm{sign}(\mathbf{U}\mathbf{V}^\top)\rangle. \tag{15}$$

To prove divergence, it is sufficient to present a matrix $\mathbf{G}$ for which $\langle \mathbf{G}, \mathrm{sign}(\mathbf{U}\mathbf{V}^\top)\rangle < 0$. In this case, $f(\mathbf{W})$ strictly increases at every step, and the algorithm perpetually moves in the wrong direction.

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1\mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103}\begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{16}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \mathrm{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{17}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle = \mathbf{u}_1^\top \mathbf{S}\mathbf{v}_1 = -\frac{43}{103}. \tag{18}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a massive singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} \approx \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{19}$$

Indeed, since $\mathbf{G} = 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}$ and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$,

$$\mathbf{G}\mathbf{v}_1 = 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 = 1001\,\mathbf{u}_1,$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{20}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{21}$$

Because $\langle \mathbf{G}, \mathbf{S}\rangle < 0$, the change in the objective function at every iteration is given by $f(\mathbf{W}_{t+1}) = f(\mathbf{W}_t) + 412.31\eta$. Thus, instead of minimizing, the algorithm strictly and monotonically diverges to $+\infty$, which completes the proof. ■

Divergence of MuonUSign

While Theorem 1 addresses the divergence of SignMuon (where the LMO is applied before sign compression), it is equally important to demonstrate that the reverse approach—MuonUSign (Algorithm 3), which applies sign compression prior to the LMO step—can also diverge in the absence of an error-feedback mechanism. Notably, the 4×4 counterexample from Theorem 1 does not cause MuonUSign to diverge because its sign matrix happens to be rank-1. However, we can construct a strict 3×3 counterexample.

Theorem 2 (Divergence of MuonUSign) *There exists a smooth function $f(\mathbf{W})$ and a matrix dimension $N_{dim} = 3$ such that the MuonUSign algorithm strictly diverges for any learning rate $\eta_t > 0$ and any momentum coefficient $\mu \in [0, 1)$, under both Standard and Nesterov momentum variants.*

Proof Consider the linear objective function $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top\mathbf{W})$. For this function, the gradient is globally constant: $\nabla f(\mathbf{W}_t) = \mathbf{G}$ for all t. Let us trace the effective direction $\tilde{\mathbf{M}}_t$ in Algorithm 3. Since $\mathbf{M}_0 = 0$, the momentum buffer at step $t \geq 1$ is simply a scaled version of the gradient:

$$\mathbf{M}_t = \sum_{i=0}^{t-1} \mu^i \mathbf{G} = \left(\frac{1-\mu^t}{1-\mu}\right)\mathbf{G} = c_t\mathbf{G}, \tag{22}$$

where $c_t > 0$ strictly, because $\mu \in [0, 1)$. Consequently, the effective direction $\tilde{\mathbf{M}}_t$ is given by:

$$\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t = c_t\mathbf{G}, & \text{(Standard momentum)} \\ \mathbf{G} + \mu\mathbf{M}_t = (1 + \mu c_t)\mathbf{G}, & \text{(Nesterov momentum)} \end{cases} \tag{23}$$

In either variant, $\tilde{\mathbf{M}}_t = \gamma_t\mathbf{G}$ for some strictly positive scalar $\gamma_t > 0$. Because the element-wise sign operator is invariant to multiplication by a strictly positive scalar, we have:

$$\mathbf{s}_t = \text{sign}(\tilde{\mathbf{M}}_t) = \text{sign}(\gamma_t\mathbf{G}) = \text{sign}(\mathbf{G}). \tag{24}$$

Thus, at every iteration, the LMO yields the exact same direction: $\mathbf{D} = A(\text{sign}(\mathbf{G}))$. The change in the objective function at step t is exactly:

$$f(\mathbf{W}_{t+1}) - f(\mathbf{W}_t) = -\eta_t\lambda_{\text{mult}}\langle\mathbf{G}, \mathbf{D}\rangle. \tag{25}$$

To prove strict divergence, we must construct a gradient $\mathbf{G}$ for which $\langle\mathbf{G}, \mathbf{D}\rangle < 0$.

Let $\mathbf{S} \in \{-1, 1\}^{3\times3}$ be the following symmetric sign matrix:

$$\mathbf{S} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}. \tag{26}$$

Because $\mathbf{S}$ is symmetric, its singular values correspond to the absolute values of its eigenvalues. The characteristic polynomial of $\mathbf{S}$ yields the exact eigenvalues $\lambda_1 = \frac{1+\sqrt{17}}{2} \approx 2.56$, $\lambda_2 = 0$, and $\lambda_3 = \frac{1-\sqrt{17}}{2} \approx -1.56$.

The largest singular value is $\sigma_1 = \lambda_1 > 0$. Because it corresponds to a strictly positive eigenvalue, the leading left and right singular vectors of $\mathbf{S}$ are identical, given by the normalized eigenvector $\mathbf{u}$. Thus, the MuonUSign LMO direction is the rank-1 positive semidefinite matrix $\mathbf{D} = \mathbf{u}\mathbf{u}^\top$.

An unnormalized eigenvector corresponding to λ_1 is analytically given by:

$$\mathbf{w} = \begin{pmatrix} -1 \\ \frac{\sqrt{17}-3}{2} \\ 1 \end{pmatrix}. \tag{27}$$

The matrix $\mathbf{w}\mathbf{w}^\top$ dictates the sign structure of $\mathbf{D}$. Let us examine its center element at index $(2, 2)$:

$$(\mathbf{w}\mathbf{w}^\top)_{2,2} = w_2^2 = \left(\frac{\sqrt{17}-3}{2}\right)^2 = \frac{13 - 3\sqrt{17}}{2}. \tag{28}$$

Since $13 = \sqrt{169}$ and $3\sqrt{17} = \sqrt{153}$, we strictly have $w_2^2 > 0$. Because $\mathbf{D} = \frac{1}{\|\mathbf{w}\|^2}\mathbf{w}\mathbf{w}^\top$, the corresponding element $D_{2,2}$ is strictly positive. Notice the crucial sign mismatch: $S_{2,2} = -1 < 0$, while $D_{2,2} > 0$.

We now define our adversarial gradient matrix $\mathbf{G}$ by assigning a massive positive weight M to this mismatched entry, and a small positive weight ϵ to all other entries, while precisely preserving the signs of $\mathbf{S}$:

$$\mathbf{G} = \epsilon\mathbf{S} - (M - \epsilon)\mathbf{e}_2\mathbf{e}_2^\top = \begin{pmatrix} \epsilon & -\epsilon & -\epsilon \\ -\epsilon & -M & \epsilon \\ -\epsilon & \epsilon & \epsilon \end{pmatrix}. \tag{29}$$

By definition, $\text{sign}(\mathbf{G}) = \mathbf{S}$, so MuonUSign deterministically selects the update direction $\mathbf{D} = \mathbf{u}\mathbf{u}^\top$ at every step, regardless of accumulated momentum.

The inner product determining the descent condition is:

$$\langle\mathbf{G}, \mathbf{w}\mathbf{w}^\top\rangle = \epsilon\langle\mathbf{S}, \mathbf{w}\mathbf{w}^\top\rangle - (M - \epsilon)w_2^2. \tag{30}$$

Since $\mathbf{S}\mathbf{w} = \lambda_1\mathbf{w}$, we have $\langle\mathbf{S}, \mathbf{w}\mathbf{w}^\top\rangle = \mathbf{w}^\top\mathbf{S}\mathbf{w} = \lambda_1\|\mathbf{w}\|^2 > 0$. Therefore:

$$\langle\mathbf{G}, \mathbf{w}\mathbf{w}^\top\rangle = \epsilon\lambda_1\|\mathbf{w}\|^2 - (M - \epsilon)\left(\frac{13 - 3\sqrt{17}}{2}\right). \tag{31}$$

Because $w_2^2 = \frac{13-3\sqrt{17}}{2} > 0$ and $\lambda_1\|\mathbf{w}\|^2 > 0$ are fixed, strictly positive constants, we can pick M to be sufficiently large (e.g., $M > \epsilon + \epsilon\lambda_1\|\mathbf{w}\|^2/w_2^2$) such that the negative term completely dominates. This makes $\langle\mathbf{G}, \mathbf{w}\mathbf{w}^\top\rangle < 0$, which immediately implies $\langle\mathbf{G}, \mathbf{D}\rangle < 0$.

Because $\langle\mathbf{G}, \mathbf{D}\rangle < 0$ and $\eta_t, \lambda_{\text{mult}} > 0$, the objective function change $-\eta_t\lambda_{\text{mult}}\langle\mathbf{G}, \mathbf{D}\rangle$ is strictly positive. Thus, at every iteration, no matter the momentum coefficient or scheme, MuonUSign takes a step in the wrong direction, causing the linear objective to strictly and monotonically diverge to $+\infty$. This completes the proof. ■

Research Hypotheses

We formulate two main hypotheses that guide the theoretical and empirical investigation of the proposed algorithms. Let $\{\mathbf{X}_t\}_{t\geq0}$ denote the sequence of global model parameters generated by the optimization algorithm, and $f(\mathbf{X})$ is the target function in federated learning setting (Equation 2).

To analyze the algorithms in both ways, we will make several assumptions.

Assumption 1 (Lower Boundedness) *The objective function $f(\mathbf{X})$ is bounded below by some value f^*, i.e., $f(\mathbf{X}) \geq f^*$ for all $\mathbf{X} \in \mathcal{X}$.*

Assumption 2 (Layer-wise Smoothness) *Following the layer-wise smoothness framework of Gruntkowska et al. (Gruntkowska et al. 2025) (Assumptions 6–9), each matrix-valued parameter $\mathbf{X}_i \in \mathbb{R}^{m_i\times n_i}$, $i \in [p]$, is equipped with the spectral norm $\|\cdot\|_{2\to2}$ and its dual, the nuclear norm $\|\cdot\|_*$. The global objective f and each local objective f_j, $j \in [N]$, are layer-wise L-smooth: for every layer $i \in [p]$ and all $\mathbf{X}, \mathbf{Y} \in \mathcal{X}$,*

$$\begin{aligned} \|\nabla_i f(\mathbf{X}) - \nabla_i f(\mathbf{Y})\|_* &\leq L_i\,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to2}, \\ \|\nabla_i f_j(\mathbf{X}) - \nabla_i f_j(\mathbf{Y})\|_* &\leq L_{i,j}\,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to2}, \end{aligned} \tag{32}$$

with layer constants $L_i, L_{i,j} \geq 0$; we denote $L := \max_{i\in[p]} L_i$. For a single matrix layer this reduces to $\|\nabla f(\mathbf{X}) - \nabla f(\mathbf{Y})\|_* \leq L\,\|\mathbf{X} - \mathbf{Y}\|_{2\to2}$. The layer-wise (L_0, L_1)-smoothness variant of (Gruntkowska et al. 2025) (Assumptions 8–9) is recovered by replacing L_i (resp. $L_{i,j}$) with $L_{0,i} + L_{1,i}\|\nabla_i f(\mathbf{X})\|_*$ (resp. $L_{0,i,j} + L_{1,i,j}\|\nabla_i f_j(\mathbf{X})\|_*$).

Assumption 3 (Stochastic Gradient) *The stochastic gradient at each client $j \in [N]$ is unbiased and has bounded variance in the dual norm:*

- *Unbiased:*

$$\mathbb{E}_{\xi\sim\mathcal{D}_j}\big[\nabla f_j(\mathbf{X};\xi)\big] = \nabla f_j(\mathbf{X}), \tag{33}$$

- *Bounded variance:*

$$\mathbb{E}_{\xi\sim\mathcal{D}_j}\big[\|\nabla f_j(\mathbf{X};\xi) - \nabla f_j(\mathbf{X})\|_*^2\big] \le \sigma^2, \quad \sigma > 0. \tag{34}$$

Remark 1 (Properties of the sign compressor) *For any $\mathbf{Y} \in \mathbb{R}^{m\times n}$ the element-wise sign operator satisfies*

$$\langle \mathbf{Y}, \mathrm{sign}(\mathbf{Y})\rangle = \|\mathbf{Y}\|_1, \qquad \|\mathrm{sign}(\mathbf{Y})\|_F^2 = \|\mathbf{Y}\|_0 \le mn, \tag{35}$$

where $\|\mathbf{Y}\|_0$ is the number of nonzero entries. In particular, $\|\mathrm{sign}(\mathbf{Y})\|_F^2 \le \beta\|\mathbf{Y}\|_1$ holds with $\beta = 1/\min_{i,j:\,Y_{ij}\neq 0}|Y_{ij}|$ whenever the nonzero entries are bounded away from zero.

Hypothesis 1 (Convergence of EF21-MuonUSign) *Under Assumptions 1–3 and 4, the Federated EF21-MuonUSign algorithm (Algorithm 7) converges in the smooth non-convex setting. There exists a choice of the learning rate η_t such that*

$$\min_{0\le t\le T} \mathbb{E}\big[\|\nabla f(\mathbf{X}_t)\|_*^2\big] = \mathcal{O}\big(T^{-1/2}\big). \tag{36}$$

The Error-Feedback mechanism restores the $\mathcal{O}(T^{-1/2})$ rate of full-precision non-Euclidean (Muon-type) methods while keeping the uplink communication at one bit per parameter. The estimator-contraction and momentum-tracking bounds of Lemmas 1–2 provide the key ingredients for this rate. In contrast, plain SignMuon does not admit such a guarantee (Theorem 1).

Hypothesis 2 (Preservation of Muon's Advantages) *Replacing the Muon LMO direction $\mathbf{D}_t = \mathbf{U}_t\mathbf{V}_t^\top$ by its sign compression $\mathrm{sign}(\mathbf{D}_t)$ preserves the core geometry of orthonormal matrix-aware updates. Consequently, on deep-learning problems SignMuon retains a training quality close to that of full-precision Muon at a $\sim 32\times$ lower uplink communication cost and outperforms the coordinate-wise SignSGD baseline in both the centralized and the federated regimes.*

Convergence Bounds for EF21-MuonUSign

Our implementation of Federated EF21-MuonUSign (Algorithm 7) uses the following steps: the server broadcasts the full model $\mathbf{X}_t$, which is equivalent to choosing the server compressor $\mathcal{C}_\downarrow^k \equiv I$ in the framework of (Gruntkowska et al. 2025). Compression is applied only on the client side, through the scaled-sign operator. This is the regime analyzed in Lemmas 7–8 of (Gruntkowska et al. 2025); below we restate the two resulting bounds in our notation. For the analysis, we adopt the exponential moving-average (EMA) momentum convention used in (Gruntkowska et al. 2025):

$$\mathbf{M}_t^{(j)} = (1-\beta)\,\mathbf{M}_{t-1}^{(j)} + \beta\,\nabla f_j(\mathbf{X}_t;\xi_t^{(j)}), \qquad \beta\in(0,1].$$

It is equivalent to the heavy-ball form $\mathbf{M}_t^{(j)} = \mu\mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ of Algorithm 7 up to a rescaling of the momentum buffer.

Assumption 4 (Contractive Compression) *The scaled-sign compressor $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}|\mathbf{Y}|\cdot\mathrm{sign}(\mathbf{Y}) = \frac{1}{d}\|\mathbf{Y}\|_1\,\mathrm{sign}(\mathbf{Y})$, where $\mathbf{Y}\in\mathbb{R}^{m\times n}$ and $d = mn$, is contractive:*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 \le (1-\alpha)\|\mathbf{Y}\|_F^2, \qquad \alpha\in(0,1],\ \forall\,\mathbf{Y}\in\mathbb{R}^{m\times n}. \tag{37}$$

For the scaled-sign operator one may take $\alpha = 1/d$, since a direct computation gives $\|\mathcal{C}(\mathbf{Y})-\mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \frac{1}{d}\|\mathbf{Y}\|_1^2$ and $\|\mathbf{Y}\|_1^2 \ge \|\mathbf{Y}\|_F^2$.

Lemma 1 (Gradient-estimator contraction) *Let Assumptions 1–3 and 4 hold, and run EF21-MuonUSign with identity compression on the server. Then the compressed gradient estimator $\mathbf{g}_t^{(j)}$ of each client $j\in[N]$ satisfies*

$$\begin{aligned}\mathbb{E}\Big[\|\mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)}\|_F^2 \,\big|\, \mathbf{X}_t, \mathbf{M}_{t-1}^{(j)}, \mathbf{g}_{t-1}^{(j)}\Big] &\le (1-\alpha)\big[(1-\beta)\,\|\mathbf{M}_{t-1}^{(j)} - \mathbf{g}_{t-1}^{(j)}\|_F^2 \\ &\quad + 2\beta\,\|\nabla f_j(\mathbf{X}_t) - \mathbf{g}_{t-1}^{(j)}\|_F^2 + 2\beta\,\sigma^2\big].\end{aligned} \tag{38}$$

This is Lemma 7 of (Gruntkowska et al. 2025) specialized to the identity server regime ($C^k \equiv I$). The estimator $\mathbf{g}_t^{(j)}$ is contracted towards the momentum $\mathbf{M}_t^{(j)}$ at a geometric rate governed by the contraction α; the identity $\mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)} = (\mathbf{M}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}) - \mathcal{C}(\mathbf{M}_t^{(j)} - \mathbf{g}_{t-1}^{(j)})$ reduces the bound to (37) together with the convexity of $\|\cdot\|_F^2$.

Lemma 2 (Momentum tracking) *Let Assumptions 1–3 and 4 hold, and let each f_j be L-smooth (Assumption 2). Then the EMA momentum tracks the true local gradient:*

$$\begin{aligned}\mathbb{E}\|\mathbf{M}_t^{(j)} - \nabla f_j(\mathbf{X}_t)\|_F^2 &\le (1-\beta)\,\mathbb{E}\|\mathbf{M}_{t-1}^{(j)} - \nabla f_j(\mathbf{X}_{t-1})\|_F^2 + \beta\,\sigma^2 \\ &\quad + C(\beta)\,L^2\,\|\mathbf{X}_t - \mathbf{X}_{t-1}\|_F^2,\end{aligned} \tag{39}$$

with $C(\beta) = (1-\beta)/\beta$ as in Lemma 8 of (Gruntkowska et al. 2025).

This is Lemma 8 of (Gruntkowska et al. 2025) (identity server regime); the last term reflects the smoothness-induced drift and is proportional to the squared length of the LMO step $\mathbf{X}_t - \mathbf{X}_{t-1}$.

Lemmas 1-2 imply that the global estimator $\mathbf{G}_t = \frac{1}{N}\sum_{j=1}^N \mathbf{g}_t^{(j)}$ remains a controlled-bias approximation of the full gradient $\nabla f(\mathbf{X}_t)$. Combined with Assumption 2, this is the key ingredient behind the $\mathcal{O}(T^{-1/2})$ rate postulated in Hypothesis 1 for our variant of EF21-MuonUSign.

Results for the smooth convex problem

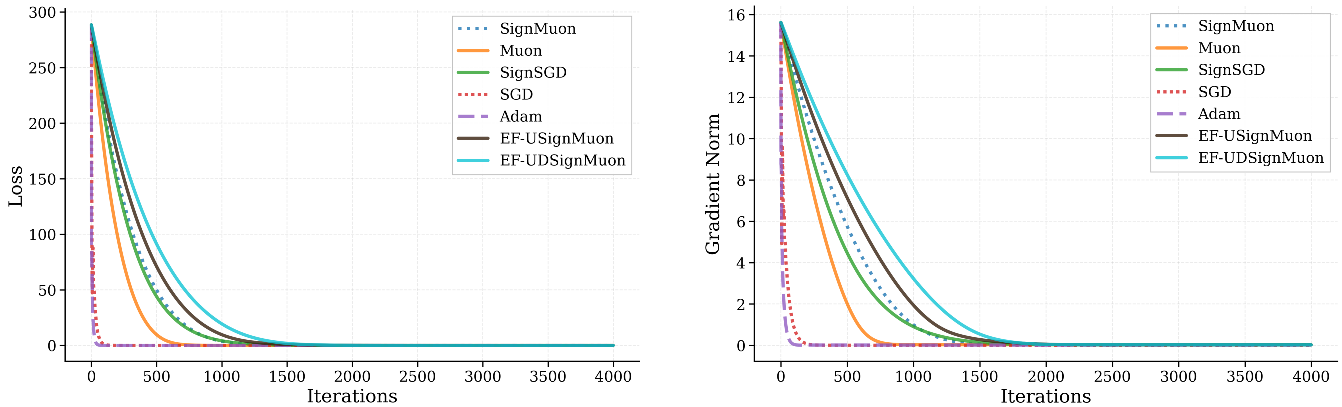
Table 3 details the grid-search ranges used for each optimizer on the synthetic problem of Equation 13, and Figure 3 shows the loss and gradient-norm dynamics.

Centralized training results

Table 4 reports the centralized CIFAR-10 results on ResNet-18, including the tuned hyperparameters and the final train/test accuracies.

Algorithm	Learning Rate (η)	Momentum (μ)
SignMuon	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Muon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF21-MuonUSign	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SignSGD	$[5 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ with step $1 \cdot 10^{-5}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SGD	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Adam	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	—

Table 3: Hyperparameter search grid for the synthetic convex problem.



(a) Loss

(b) Frobenius norm of the gradient

Figure 3: Smooth convex problem (Equation 13 for 500×500).

Dataset	Optimizer	Epoch	Batch	LR	LR Aux	Mom	Train Acc	Test Acc
CIFAR-10	SignMuon	75	128	0.001	0.0001	0.9	99.69%	92.55%
	Muon			0.015	0.001		99.94%	93.70%
	SGD						99.98%	93.93%
	SignSGD			0.001	0.0001		93.19%	89.11%
	Adam						99.77%	93.03%
	EF21-MuonUSign			0.008	0.001		99.96%	93.82%

Table 4: CIFAR-10: Centralized Learning on ResNet-18.

Federated training results. Experiment 1

Table 5 and Figure 4 report the scalability of SignMuon with respect to the number of clients (Experiment 1).

Federated training results. Experiment 2

Table 6 and Figure 5 report the comparison of optimizers with $N = 3$ clients (Experiment 2).

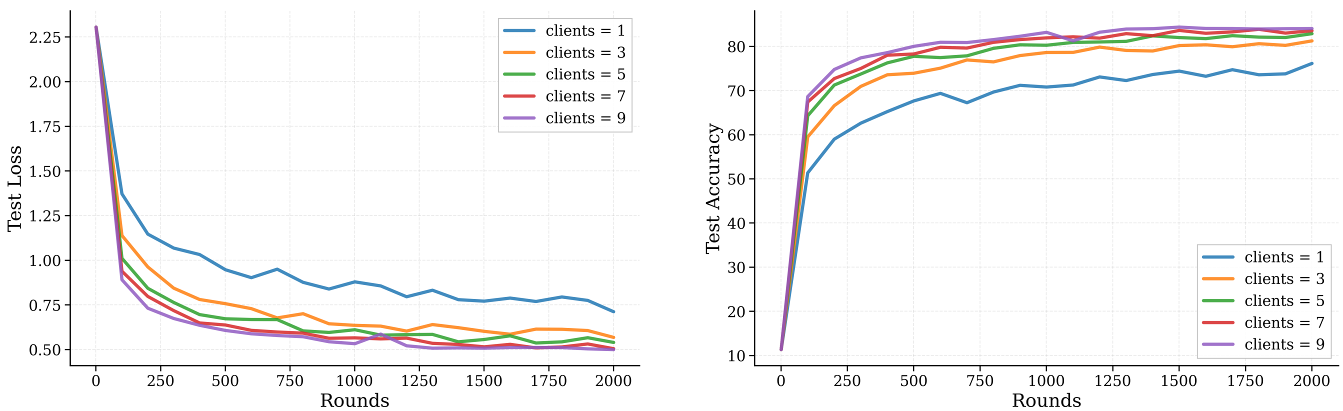
Federated training results. Experiment 3

Figure 6 (together with Table 2 in the main text) reports the comparison of optimizers with $N = 10$ clients (Experiment 3).

Algorithms

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	1	1	64	0.001	0.9	76.13%
			3					81.24%
			5					82.87%
			7					83.54%
			9					84.03%

Table 5: CIFAR-10 Federated Learning on CNN2. Experiment 1. Scalability of the SignMuon Algorithm.



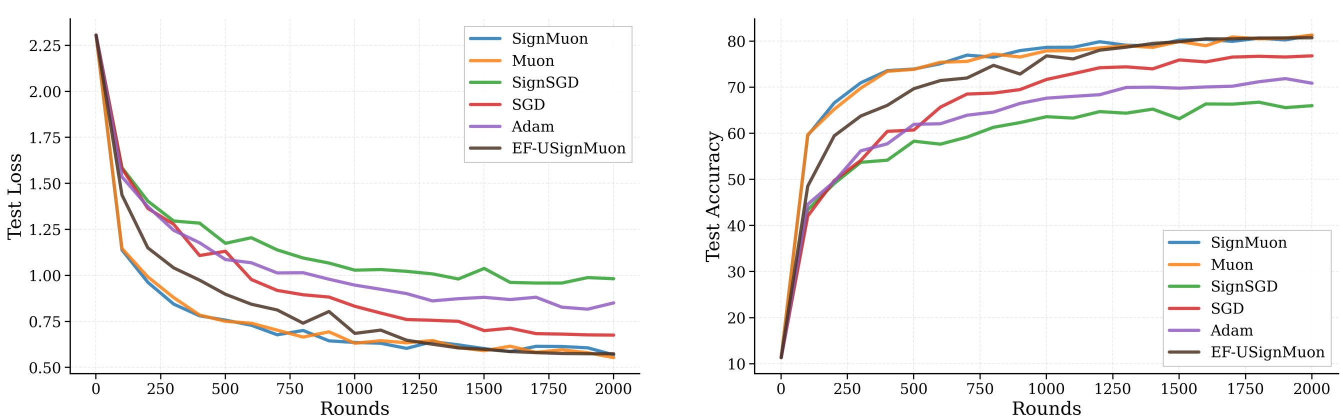
(a) Test Loss

(b) Test Accuracy

Figure 4: CIFAR-10 Federated Learning on CNN2. Experiment 1. Scalability of the SignMuon Algorithm.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	3	1	64	0.001	0.9	81.24%
	Muon					0.05		81.27%
	SignSGD					0.001		65.96%
	SGD					0.03		76.77%
	Adam					0.001		70.83%
	EF21-MuonUSign					0.02		80.69%

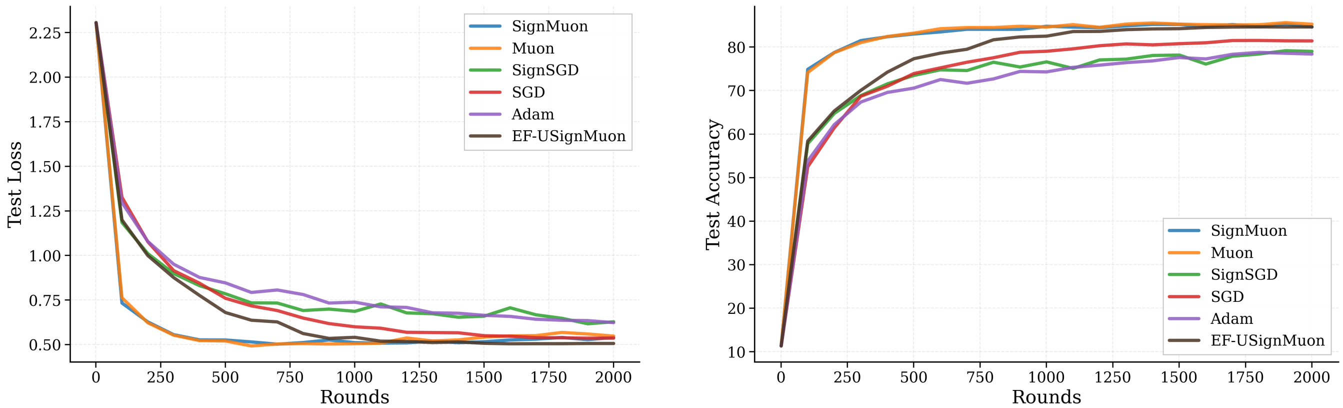
Table 6: CIFAR-10 Federated Learning on CNN2. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.



(a) Test Loss

(b) Test Accuracy

Figure 5: CIFAR-10 Federated Learning on CNN2. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.



(a) Test Loss

(b) Test Accuracy

Figure 6: CIFAR-10 Federated Learning on CNN2. Experiment 3: Comparison of Optimization Algorithms with (N = 10) Clients.

Algorithm 1: MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$; iteration count ns_steps
Output: Muon LMO direction $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

```
Y ← ReshapeTo2D(Y)                      ▷ Flatten tensor to matrix m × n
Y ← Y/‖Y‖_F                             ▷ Normalize (optional)
for k = 1 to ns_steps do                ▷ 5th-order Newton-Schulz orthogonalization
    A ← Y Y^⊤
    Y ← a Y − b A Y + c A^2 Y
end for
D ← ReshapeToOriginal(Y)                ▷ Reshape back to the original tensor shape
return D
```

Algorithm 2: Centralized SignMuon

Input: Initial model $\mathbf{X}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ, learning rate η_t, $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations ns_steps
Output: Updated model $\mathbf{X}$

```
M_0 ← 0
for t = 0 to (T − 1) do
    Stochastic gradient G_t
    M_t ← μ M_{t−1} + G_t               ▷ Momentum accumulation
    M̃_t = { G_t + μ M_t,  (Nesterov momentum),
          { M_t,          (otherwise)
    D_t ← MuonLMO(M̃_t)                  ▷ Algorithm 1
    s_t ← sign(D_t)                     ▷ Sign compression
    X_{t+1} ← X_t − η_t s_t             ▷ Update parameters
end for
```

Algorithm 3: Centralized MuonUSign

Input: Initial model $\mathbf{X}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ, learning rate η_t, $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations ns_steps
Output: Updated model $\mathbf{X}$

```
M_0 ← 0
for t = 0 to (T − 1) do
    Stochastic gradient G_t
    M_t ← μ M_{t−1} + G_t               ▷ Momentum accumulation
    M̃_t = { G_t + μ M_t,  (Nesterov momentum),
          { M_t,          (otherwise)
    s_t ← sign(M̃_t)                     ▷ Sign compression BEFORE LMO
    D_t ← MuonLMO(s_t)                  ▷ Algorithm 1
    X_{t+1} ← X_t − η_t λ_mult D_t      ▷ Update parameters
end for
```

Algorithm 4: Centralized EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ, learning rate η_t, $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations ns_steps
Output: Updated model $\mathbf{X}$

```
M_0 ← 0,   g_0^est ← 0
for t = 0 to (T − 1) do
    Stochastic gradient G_t
    M_t ← μ M_{t−1} + G_t
    Δ_t ← M_t − g_t^est;   α_t ← mean(|Δ_t|);
    g_{t+1}^est ← g_t^est + α_t sign(Δ_t)   ▷ Uplink EF21 estimator (1-bit residual)
    D_t ← MuonLMO(g_{t+1}^est)              ▷ Algorithm 1
    X_{t+1} ← X_t − η_t λ_mult D_t
end for
```

Algorithm 5: Centralized EF21-MuonUDSign (bidirectional)

Input: Initial model $\mathbf{X}_0 = \mathbf{W}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ, learning rate η_t, $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations ns_steps
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0$
for $t = 0$ to $(T-1)$ **do**
  Stochastic gradient $\mathbf{G}_t$ at the broadcast model $\mathbf{W}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + \mathbf{G}_t$ ▷ Momentum
  $\Delta_t^{\uparrow} \leftarrow \mathbf{M}_t - \mathbf{g}_t^{\text{est}}$
  $\alpha_t^{\uparrow} \leftarrow \text{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_{t+1}^{\text{est}} \leftarrow \mathbf{g}_t^{\text{est}} + \alpha_t^{\uparrow}\,\text{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21 estimator (1-bit)
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{g}_{t+1}^{\text{est}})$ ▷ Algorithm 1
  $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \lambda_{\text{mult}} \mathbf{D}_t$
  $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_{t+1} - \mathbf{W}_t$
  $\alpha_t^{\downarrow} \leftarrow \text{mean}(|\Delta_t^{\downarrow}|)$
  $\mathbf{W}_{t+1} \leftarrow \mathbf{W}_t + \alpha_t^{\downarrow}\,\text{sign}(\Delta_t^{\downarrow})$ ▷ Downlink EF (1-bit model increment)
end for

Algorithm 6: Federated SignMuon

Input: Initial model $\mathbf{X}_0$, clients number N, rounds number rounds, local steps number n_steps, learning rate for the main layers η_t, learning rate for the last layer η_t^{aux}, momentum coefficient μ
Output: Updated global model $\mathbf{X}$

for $j = 1$ to N **do**
  $\mathbf{M}_0^{(j)} \leftarrow 0$ ▷ Initialization of local buffers on clients
end for
Initialize AdamW for last_layer with learning rate η_0^{aux}
for $t = 0$ to $(\text{rounds} - 1)$ **do**
  for $j = 1$ to N **in parallel do**
    $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$ ▷ Retrieve global model
    for $k = 1$ to n_steps **do**
      Sample a batch $\xi_{t,k}^{(j)}$
      $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$
      $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$
    end for

    $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \text{n\_steps}$ ▷ Gradient averaging
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ ▷ Local momentum update
    $\mathbf{D}_t^{(j)} \leftarrow \text{MuonLMO}(\mathbf{M}_t^{(j)})$ ▷ Algorithm 1
    $\mathbf{s}_t^{(j)} \leftarrow \text{sign}\big(\mathbf{D}_t^{(j)}\big)$ ▷ Sign compression

    Send $\mathbf{s}_t^{(j)}$ to server
    for each $\mathbf{W}^{\text{last}}$ in last_layer do
      Send $\mathbf{G}_t^{(j)}$ to the server ▷ Raw gradient for the head
    end for
  end for
  $\mathbf{s}_t^{\text{agg}} \leftarrow \text{sign}\Big(\sum_{j=1}^{N} \mathbf{s}_t^{(j)}\Big)$ ▷ Majority vote (matrix layers)
  $\bar{\mathbf{G}}_t^{\text{last}} \leftarrow \frac{1}{N} \sum_{j=1}^{N} \mathbf{G}_t^{(j)}$ ▷ Average raw gradient for the head
  $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}$ ▷ Matrix parameter update

  $\mathbf{X}_{t+1}^{\text{last}} \leftarrow \text{AdamW}(\mathbf{X}_t^{\text{last}}, \bar{\mathbf{G}}_t^{\text{last}}, \eta_t^{\text{aux}})$ ▷ AdamW on the averaged gradient
end for
return $\mathbf{X}_{\text{rounds}}$

Algorithm 7: Federated EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, clients number N, rounds number rounds, local steps number $\mathrm{n\_steps}$, learning rate for the main layers η_t, learning rate for the last layer η_t^{aux}, momentum coefficient μ

Output: Updated global model $\mathbf{X}$

$\mathbf{G}_0 \leftarrow 0$ ▷ Initialize the global gradient estimate on the server (for the main layers)
for $j = 1$ to N **do**
  $\mathbf{g}_0^{(j)} \leftarrow 0,\ \mathbf{M}_0^{(j)} \leftarrow 0$ ▷ Initialize local estimators and momentum
end for
Initialize AdamW for last_layer with learning rate η_0^{aux}
for $t = 0$ to $(\mathrm{rounds} - 1)$ **do**
  for $j = 1$ to N **in parallel do** ▷ Local step of EF21
    $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$ ▷ Retrieve global model
    $\mathbf{G}_t^{(j)} \leftarrow 0$
    for $k = 1$ to $\mathrm{n\_steps}$ **do**
      Sample a batch $\xi_{t,k}^{(j)}$
      $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$
      $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$
    end for
    $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \mathrm{n\_steps}$ ▷ Gradient averaging
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ ▷ Local momentum update
    for each parameter $\mathbf{W}$ **not in last_layer do**
      $\Delta_t^{(j)} \leftarrow \mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)}$ ▷ Difference computation
      $\alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$ ▷ Adaptive scaling factor
      $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\Delta_t^{(j)})$ ▷ Compress shifted gradient
      $\mathbf{g}_{t+1}^{(j)} \leftarrow \mathbf{g}_t^{(j)} + \alpha_t^{(j)} \mathbf{s}_t^{(j)}$ ▷ Compute gradient estimator
      Send $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$ to the server
    end for
    for each parameters $\mathbf{W}^{\text{last}}$ **in last_layer do**
      Send $\mathbf{G}_t^{(j)}$ to the server
    end for
  end for
  $\mathbf{G}_{t+1} \leftarrow \mathbf{G}_t + \frac{1}{N} \sum_{j=1}^{N} \alpha_t^{(j)} \mathbf{s}_t^{(j)}$ ▷ Update the global estimator on the server
  for each parameters $\mathbf{X}$ **do**
    $\mathbf{D}_{t+1} \leftarrow \mathrm{MuonLMO}(\mathbf{G}_{t+1})$ ▷ Algorithm 1
    $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{D}_{t+1}$ ▷ Update the matrix parameters
  end for
  $\nabla^{\text{last}} \leftarrow \frac{1}{N} \sum_{j=1}^{N} \mathbf{G}_t^{(j)}$
  $\mathbf{X}_{t+1}^{\text{last}} \leftarrow \mathrm{AdamW}(\mathbf{X}_t^{\text{last}}, \nabla^{\text{last}}, \eta_t^{\text{aux}})$ ▷ Classifier update
end for
return $\mathbf{X}_{\mathrm{rounds}}$

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